CYBER IMMUNITY GRID

SELF HEALING SMART NETWORK

Team Information:

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Abstract

Cyber threats are increasing rapidly, while traditional networks mainly rely on threat detection and manual recovery. Cyber Immunity Grid (CIG) is a web-based simulation inspired by the human immune system, designed to demonstrate a self-healing smart network.

CIG visualizes devices, attack severity, infection spread, and automated healing through an interactive dashboard, demonstrating autonomous detection, isolation, and recovery to improve network resilience.

Problem Domain

- Cyber-attacks in modern networks are rapidly increasing, challenging traditional security systems.
- Existing cybersecurity solutions focus on detection but lack automated recovery and self-healing.
- Infected devices can spread attacks across networks, causing widespread disruption.
- Lack of visual tools to understand attack impact and healing.
- A unified cyber-immunity platform is needed to automate threat detection, isolation, and recovery.

Motivation & Objectives

- CIG provides a self-healing smart network, inspired by the human immune system.
- It provides real-time visibility of device health, infection levels, and recovery through an interactive user interface.
- The system enables automated threat detection, device isolation, and healing processes without manual intervention.
- CIG offers a centralized security view to simplify continuous network monitoring and response.

Literature Review

Table 1: Literature Rivew

Platform	Limitations	Findings
Cybersecurity Dashboards [1]	• Focus mainly on monitoring • Manual response required	• Visualization improves situational awareness • Incentives boost learning
Cyber Threat Observatory [2]	• No automated recovery. • Reactive threat handling.	• Centralized threat tracking enhances detection • Real-time alerts reduce response delay
Threat Intelligence Dashboards [3]	• Threat Intelligence Dashboards. • Limited device-level recovery.	• Correlating threat data improves analysis. • Automation increases system resilience
Traditional Network Security Systems[4]	• Detection-centric approach. • No isolation or healing automation.	• Early detection possible. • Manual recovery increases downtime.

Methodology

Prototype Design (UI/UX Design)

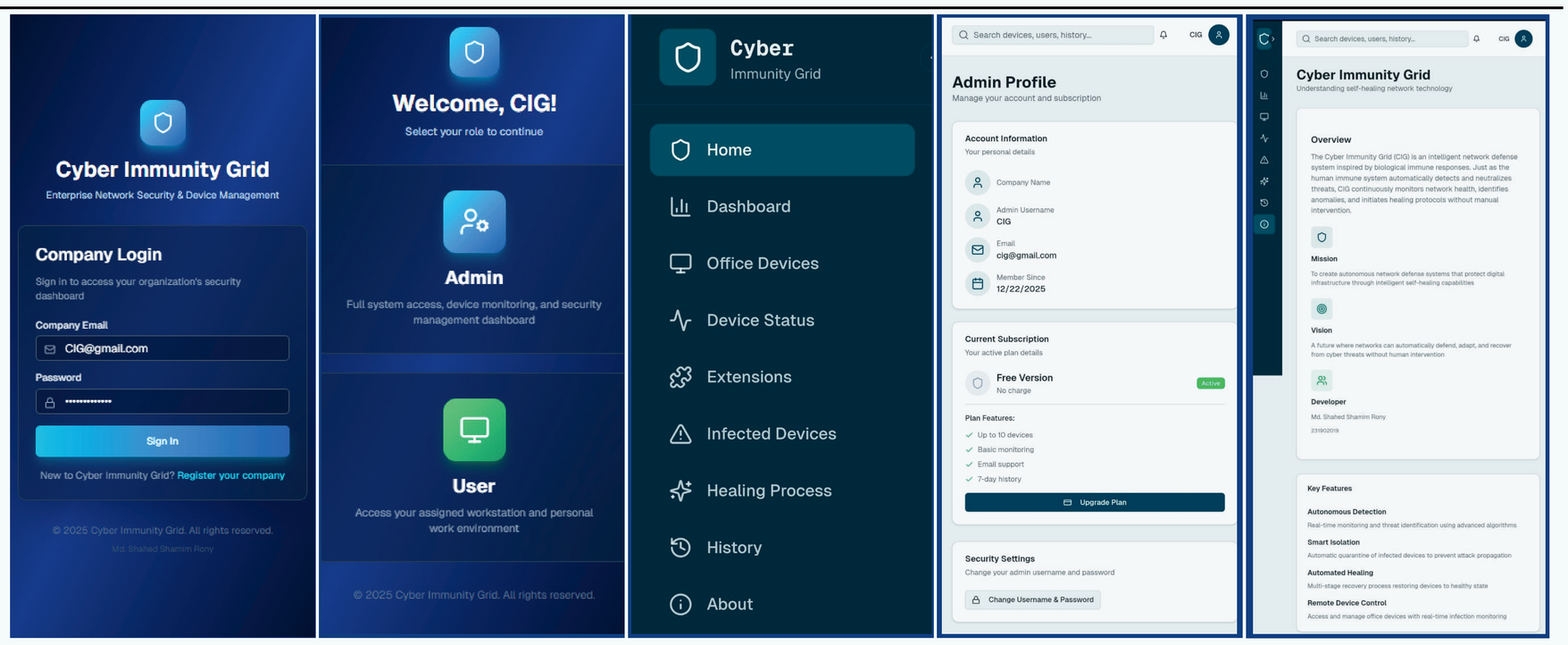


Fig 1: UI/UX of Cyber Immunity Grid

SRS

Functional Requirements

- Provide real-time display of connected devices and their health status.
- Detect cyber threats and monitor infection spread across the network.
- Automatically initiate device isolation and healing processes.
- Provide continuous updates on device recovery and system status.
- Maintain logs of infection events and recovery history.

Non-Functional Requirements

- Ensure fast system response for monitoring and recovery actions.
- Maintain high reliability and consistent system performance.
- Provide an intuitive and easy-to-use admin interface.
- Support scalability for future feature and device expansion.
- Ensure secure access and controlled administrative operations

SDLC

Table 2: SDLC Model Selection

Priority Criteria	Priority Value	Waterfall	V-Model	Iterative	Spiral	Agile	Prototype
Complexity Handling	5	No	Yes	Yes	Yes	Yes	Yes
Scalability & Flexibility	5	No	No	Yes	Yes	Yes	Yes
Security & Reliability	5	Yes	Yes	No	Yes	Yes	No
User Engagement & Feedback Integration	5	No	No	Yes	Yes	Yes	Yes
Development Speed	5	No	No	Yes	No	Yes	Yes
Risk Management	5	No	No	No	Yes	Yes	No
Weighted Total Score		5	10	20	25	30	20

Selected Model : Agile

DFD

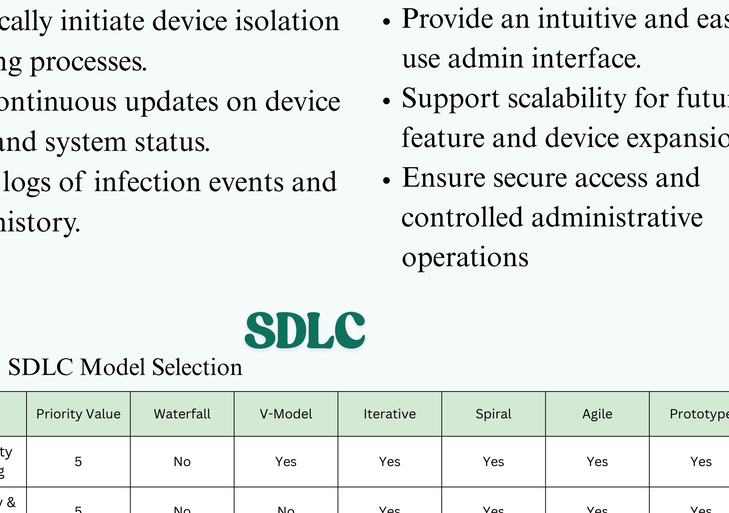


Fig 2: DFD of Cyber Immunity Grid

Use Case Diagram

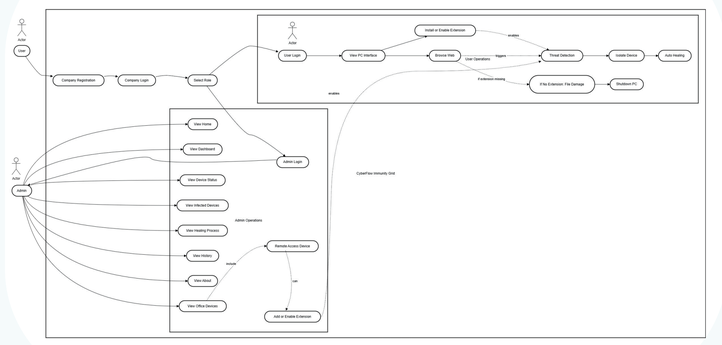


Fig 3: Use Case Diagram of Cyber Immunity Grid

Sequence Diagram

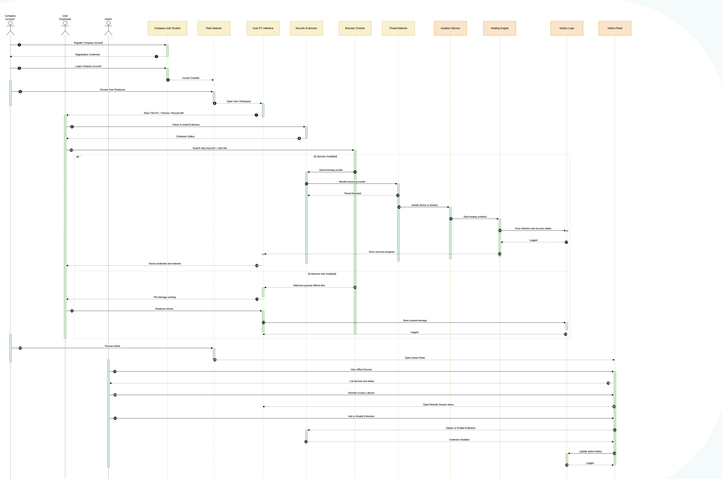


Fig 4: Sequence Diagram of Cyber Immunity Grid

Class Diagram

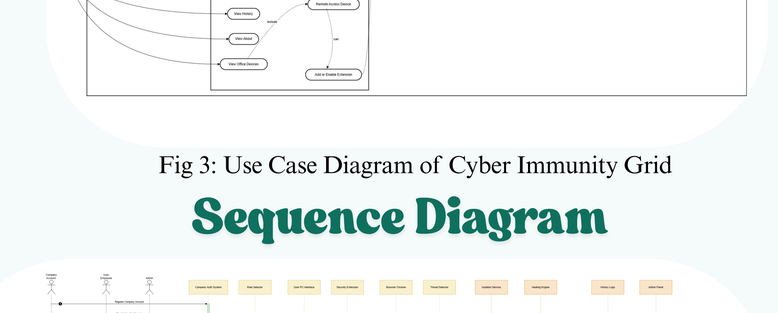


Fig 5: Class Diagram of Cyber Immunity Grid

Social Impact

- Improves understanding of cyber-resilience.
- Promotes automated security concepts.
- Faster cyber Incident Response and Recovery for commercial companies.
- Foundation for future AI-based cyber defense systems.

Conclusion

Cyber Immunity Grid (CIG) is an enterprise security platform that goes beyond detection by automatically isolating infected devices and triggering self-recovery. It minimizes business disruption, reduces human dependency, and provides real-time visibility into threats and device health through a centralized dashboard—strengthening operational efficiency and network resilience.

Reference

[1] H. Zhao and B. Silverajan, Evaluating Cybersecurity Dashboards, ARES, 2024.
[2] M.-A. Kaufhold et al., Cyber Threat Observatory, ECIS, 2022.
[3] G. Madhan et al., Cyber Threat Intelligence Dashboard, IJRD, 2025.
[4] https://www.gtt.net/resources/blog/nsaas-vs-traditional-security-solutions/